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Robust Color Image Watermarking Using Geometric Invariant Quaternion Polar Harmonic Transform

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It is a challenging work to design a robust color image watermarking scheme against geometric distortions. Moments and moment invariants have become a powerful tool in robust image watermarking owing to their image description capability and geometric invariance property. However, the existing moment-based watermarking schemes were mainly designed for gray images but not for color images, and detection quality and robustness will be lowered when watermark is directly embedded into the luminance component or three color channels of color images. Furthermore, the imperceptibility of the embedded watermark is not well guaranteed. Based on algebra of quaternions and polar harmonic transform (PHT), we introduced the quaternion polar harmonic transform (QPHT) for invariant color image watermarking in this article, which can be seen as the generalization of PHT for gray-level images. It is shown that the QPHT can be obtained from the PHT of each color channel. We derived and analyzed the rotation, scaling, and translation (RST) invariant property of QPHT. We also discussed the problem of color image watermarking using QPHT. Experimental results are provided to illustrate the efficiency of the proposed color image watermarking against geometric distortions and common image processing operations (including color attacks).

Categories and Subject Descriptors: K.6.5 [Management of Computing and Information Systems]: Security and Protection; I.4.9 [Image Processing and Computer Vision]: Applications

General Terms: Algorithms, Management, Security

Additional Key Words and Phrases: Color image watermarking, quaternion polar harmonic transform, geometric invariance, robustness, imperceptibility

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1. INTRODUCTION

Because of the rapid advance of network technology, humans can arbitrarily and easily access or distribute any multimedia data from networks. Hence, the protection of intellectual property becomes more and more attentive and important for the society. Digital watermarking has been proposed as a technology to ensure multimedia

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copyright protection by embedding an imperceptible, yet detectable signal in digital multimedia content such as images or video [Lin et al. 2011]. For different purposes, digital watermarking can be branched into two classifications: robust watermarking technique and fragile watermarking technique. Robust digital watermarking is used to protect ownership of the digital media. In contrast, the purpose of fragile watermarking technique is digital media authentication, that is, to ensure the integrity of the digital media [Cheddad et al. 2010; Kumar et al. 2011].

In the past decade, researchers have made the great efforts in developing robust image watermarking algorithms. On the other hand, attacks against image watermarking systems have become more sophisticated. In general, these attacks on image watermarking systems can be categorized into noise-like common image processing operations and geometric distortions. While the noise-like common image processing operations reduce watermark energy, geometric distortions induce synchronization errors between the original and the embedded watermark pattern and therefore can mislead the watermark detector [Zheng et al. 2009, 2007].

Until now, most of the previous image watermarking is mainly robust against common image processing operations, but show severe problems to geometric distortions. Fortunately, several approaches that counterattack geometric distortions have been developed. These schemes can be roughly divided into Spread spectrum modulation, Template insertion, Feature-based embedding, and Invariant domain [Kumar et al. 2011; Zheng et al. 2009, 2007].

Spread Spectrum Modulation is probably the most popular approach for data hiding, which spreads the digital watermark over the host image. Spread spectrum embedding could be implemented by two main ways, namely, additive and multiplicative spread spectrum embedding [Cox et al. 1997; Zhong et al. 2006]. The additive spread spectrum scheme uniformly spreads the watermark bit over the host image while the multiplicative spread spectrum spreads the watermark bit according to the host contents. Since the original host image is generally not available at the watermark decoder side, blind decoders are usually employed. Its robustness against common image processing operations and some geometric distortion, and its simple decoder structure make spread spectrum attractive for image watermarking [Valizadeh et al. 2011]. But the interference effect of the host image, which causes the watermark decoding performance degradation, is usually a major concern of the spread spectrum modulation [Valizadeh et al. 2012].

Template Insertion. Another strategy for detecting watermarks after geometric distortion is to identify what the distortions are, and apply the inverse transform before watermark extraction. This can be done by embedding a template along with the watermark into the host image [Zheng et al. 2009; Barni 2005]. The researchers proposed to embed two watermarks, a template and a spread spectrum message containing the information or payload. The template contains no information itself, but is used to detect the transformations undergone by the watermarked image. This type of template is applicable for all images. However, they can also be easily removed [Liu et al. 2012] since they usually represent peaks in a transform domain. This fact may increase collusion attempts to discern the registration pattern and, once found, the registration pattern could be removed from the watermarked image, thus restricting the invertible ability of any geometric distortions [Zhao et al. 2012].

Feature-Based Embedding is a new resynchronization technique, which uses image content to recover the watermark after geometrical transformations [Seo et al. 2006]. Its basic idea is that, by binding the watermark with the geometrically invariant image features, the watermark detection can be done without synchronization error [Wang et al. 2007]. Through experiments, we find out that the feature points often cannot be located accurately, if the watermarked image goes through common processing and

geometrical transforms. In addition, watermark capacity is often limited, because the watermark is embedded into the feature based on local regions [Yuan et al. 2013; Gao et al. 2010; Tsai et al. 2011].

Invariant Domain. The obvious way to achieve resilience against geometric distortions is to use an invariant domain. In several works [Simitopoulos et al. 2003; Lai et al. 2010; Ali et al. 2014; Coatrieux et al. 2013; Su et al. 2013], the watermark was embedded in an affine-invariant domain by using Fourier-Mellin transform, generalized Radon transform, singular value, QR decomposition, and histogram shape, respectively. Once the Fourier-Mellin transform, generalized Radon transform, singular value, QR decomposition, and histogram technique are applied to an image, the image will be transformed to the rotation, scaling, and translation (RST) invariant domain. The watermark embedded into this domain can be RST invariant. However, the implementation difficulties hinder the research of watermarking schemes based on this principle.

More recently, moment and moment invariants have found extensive applications in the field of image watermarking [Papakostas et al. 2012]. Image moments are efficient image content descriptors, which have the advantage to fully reconstruct the initial image after the embedment of the watermark information. Moreover, the invariant properties of the moments to remain unchanged under common geometric transformations significantly increase the robustness of the watermarks in such kind of attacks. In 2000, Alghoniemy et al. [2000] first incorporated image moments in watermarking systems, and applied seven Hu moments invariants in watermark embedding so as to achieve robustness to RST attacks. Since then, numerous research efforts, which are reported in the literature [Papakostas et al. 2012], applied image moments for watermark insertion including also new moment families with improved description capabilities. Kim et al. [2003] introduced a robust image watermark based on normalized Zernike moments, in which the watermark signal is designed with Zernike moments and the signal is added to the cover image in the spatial domain after the reconstruction process. Xin et al. [2007] proposed an image watermarking method by using *Zernike moments* (ZMs) and *pseudo-Zernike moments* (PZMs). This method introduced the bit sequence watermark insertion via dither modulation that many researchers adopted later in their works [Papakostas et al. 2012]. Wang et al. [2010] proposed a robust digital image watermarking scheme based on PZMs and image normalization. Singhal et al. [2009] proposed a robust image watermarking algorithm using local Zernike moments, which are computed over circular patches around feature points. The proposed algorithm locally computes Zernike moments and modifies them to embed watermarks, achieving robustness against cropping and local geometric distortions. In the scheme of [Wang et al. 2010], watermark is inserted into the perceptually significant Krawtchouk moment invariants of the original image, and watermarking based on Krawtchouk moment invariants are local, that is, the embedded watermark affects only a selected portion of the original image, the position of which can be decided by the user. Singh et al. [2013] introduced a novel image adaptive technique for high capacity watermarking scheme using accurate and fast *radial harmonic Fourier moments* (RHFMs). The high embedding capacity is achieved by improving the hiding ratio after reducing inaccuracies in the computation of RHFMs. Zhang et al. [2011] proposed a new watermarking approach which allows watermark detection and extraction under affine transformation attacks. The novelty of the approach stands on a set of affine invariants derived from Legendre moments. Watermark embedding and detection are directly performed on this set of invariants. Yuan et al. [2011] proposed a local Zernike moments based watermarking scheme where the watermarked image/region can be obtained directly by inverse Zernike transform. Based on moment normalization and radial symmetry transform, Nikolaidis [2012] proposed an affine transformation

invariant image watermarking method against geometrical distortions. Elshoura et al. [2013] examined the problem of noisy image reconstruction by the method of orthogonal moments, and presented an empirical comparative study of Tchebichef and Zernike moments in image watermarking applications. Venkataramana et al. [2007] proposed an RST robust watermarking method that uses Krawtchouk moment invariants for watermark construction. The novelty of this method lies on the usage of the moment invariants in the detection stage, while the insertion is performed in the moment space and the watermarked image is computed by reconstruction. Zhu et al. [2010] first presented an approach of RST invariant analysis for images. This approach achieves a set of completely invariant descriptors from the complex moments of the original image's Radon projection. Then, they proposed a watermarking scheme which can resist global geometric transforms. Savelonas et al. [2010] employed Fourier coefficients for moment-based image analysis and the fractional Fourier transformation for watermark embedding. This approach maintains the advantages of spatial and frequency domain representations, offers two additional degrees of freedom associated with the fractional Fourier transformation angles, whereas it provides increased detectability of the embedded watermarks at the received end.

Through in-depth research and analysis, we can see that the moment-based image watermarking is one kind of effective watermarking schemes. However, there are some main drawbacks indwelled in current moment-based image watermarking schemes, which will affect inevitably the image watermarking performances. First, the existing moment-based watermarking schemes were mainly designed for gray images but not for color images. However, color information usually plays a vitally important role in the world. If the watermark is directly embedded into the luminance component or three color channels of color images [Chou et al. 2010; Niu et al. 2011], the watermark detection quality and robustness will be lowered because of ignoring the correlation between different color channels. Second, the imperceptibility of the embedded watermark is not well guaranteed, this is because that the kernel computation of image moments involves computation of a number of factorial terms, which inevitably cause the numerical stability of these moments. For example, the numerical stability of ZMs/PZMs will break down when the number of moments is increased up to a certain point.

Based on a set of orthogonal projection bases, Yap et al. [2010] introduced a set of 2D transforms named *Polar Harmonic Transform* (PHT) to generate a set of features which are geometric invariant in 2010. Unlike the conventional image moments such as ZMs/PZMs/RHFMs, the kernel computation of PHT is extremely simple and has no numerical stability issue whatsoever. This implies that PHT encompass the orthogonality and invariance advantages of ZMs/PZMs/RHFMs, but are free from their inherent limitations. As a result, we believe PHT coefficients are more suitable for robust image watermarking.

Based on algebra of quaternions and PHT, we proposed a robust color image watermarking algorithm using geometric invariant *quaternion polar harmonic transform* (QPHT), which can be seen as the generalization of PHT for gray-level images. The novelty of the proposed algorithm includes (1) QPHT is introduced, and the invariant property of QPHT is analyzed and discussed; (2) the accurate QPHT coefficients selection strategy is discussed; and (3) different from existing invariant domain watermarking techniques, the proposed QPHT-based approach embed watermark into host color image holistically as a vector field and can achieve good imperceptibility and robustness.

The rest of this article is organized as follows. Section 2 recalls some preliminaries about the quaternions and PHT. In Section 3, the definition of QPHT is presented. Section 4 analyzes the *rotation, scaling, and translation* (RST) invariant property of QPHT. Section 5 discusses the robust color image watermarking using QPHT.

Simulation results in Section 6 will show the performance of our scheme. Finally, Section 7 concludes this presentation. Appendices A gives the relationship between the QPHT and the conventional PHT for single channel of the RGB image, and Appendices B gives the relationship between the inverse QPHT and the conventional inverse PHT for single channel of the RGB image, respectively.

2. PRELIMINARIES

2.1. Quaternion Representation of A Color Image

Quaternions, a generalization of the complex numbers, were introduced by the mathematician Hamilton in 1843 [Kantor 1989]. A quaternion consists of one real part and three imaginary parts as follows:

$$q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \quad (1)$$

where a, b, c , and d are real numbers, and $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ are three imaginary units obeying the following rules

$$\begin{aligned} \mathbf{i}^2 + \mathbf{j}^2 + \mathbf{k}^2 &= -1 \\ \mathbf{ij} &= -\mathbf{ji} = \mathbf{k}, \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \mathbf{ki} = -\mathbf{ik} = \mathbf{j}. \end{aligned} \quad (2)$$

The conjugate and modulus of a quaternion are respectively, defined by

$$q^* = a - b\mathbf{i} - c\mathbf{j} - d\mathbf{k}, \quad |q| = \sqrt{a^2 + b^2 + c^2 + d^2}. \quad (3)$$

Using this representation, a color image $f(x, y)$, sized by $M \times N$, can be considered as an array of pure quaternion numbers (e.g., with no real parts)

$$f(x, y) = f_R(x, y)\mathbf{i} + f_G(x, y)\mathbf{j} + f_B(x, y)\mathbf{k}, \quad (4)$$

where $f_R(x, y), f_G(x, y)$, and $f_B(x, y)$ represent respectively the classical Red, Green, and Blue components, and $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ are the complex operators.

Obviously, by using the quaternion-type representation, a color image can be treated as a vector field and be processed directly, without losing color information.

2.2. Polar Harmonic Transforms

Yap et al. [2010] introduced a set of 2D transforms named Polar Harmonic Transform (PHT) based on a set of orthogonal projection bases. Compared with other orthogonal moment, PHT has a better image reconstruction, lower noise sensitivity, and lower computational complexity. Besides, the PHT is free of numerical instability issues so that high order moments can be obtained accurately.

The PHT coefficients $M_{n,l}$ of order n with repetition l (n and l in (5) are generally called order and repetition, respectively), $|n| = |l| = 0, 1, \dots, \infty$, is defined as

$$M_{n,l} = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) [H_{n,l}(r, \theta)]^* r dr d\theta, \quad (5)$$

where $[\cdot]^*$ denotes the complex conjugate and the basis $H_{n,l}(r, \theta)$ can be decomposed into radial and circular components

$$H_{n,l}(r, \theta) = R_n(r) e^{il\theta}, \quad (6)$$

with the radial kernel being a complex exponential in the radial direction

$$R_n(r) = e^{i2\pi nr^2} \quad (7)$$

and satisfying orthogonality condition

$$\int_0^1 R_n(r) [R_m(r)]^* r dr = \frac{1}{2} \delta_{n,m}. \quad (8)$$

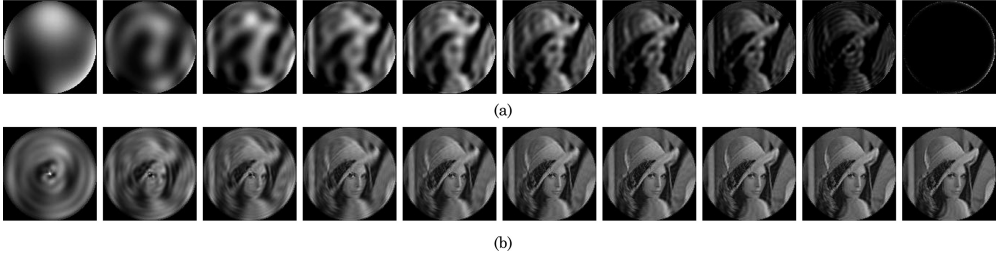


Fig. 1. A sample of reconstructed images (moment orders $K = 5, 10, 15, 20, 25, 30, 35, 40, 45, 50$): (a) ZMs for standard gray image “Lena,” (b) PHT for standard gray image “Lena.”

and also

$$\int_0^{2\pi} \int_0^1 H_{n,l}(r, \theta) [H_{n',l'}(r, \theta)]^* r dr d\theta = \pi \delta_{n,n'} \delta_{l,l'}, \quad (9)$$

where π is the normalization factor, $[H_{n',l'}(r, \theta)]^*$ is the conjugate of $H_{n',l'}(r, \theta)$, and $\delta_{n,n'}$ and $\delta_{l,l'}$ are the Kronecker symbols defined as

$$\delta_{i,j} = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases}.$$

The image (gray image) $f(r, \theta)$ can be decomposed with the set of $H_{n,l}(r, \theta)$ as

$$f(r, \theta) = \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l} R_n(r) \exp(il\theta), \quad (10)$$

where $M_{n,l}$ is the PHT coefficients of order n with repetition l .

Following the principle of orthogonal function, the image function $f(r, \theta)$ can be reconstructed approximately by limited orders of PHT coefficients ($n \leq n_{\max}$, $l \leq l_{\max}$). The more orders used, the more accurate the image description

$$f'(r, \theta) = \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l} R_n(r) \exp(il\theta) \approx \sum_{n=-n_{\max}}^{n_{\max}} \sum_{l=-l_{\max}}^{l_{\max}} M_{n,l} R_n(r) \exp(il\theta), \quad (11)$$

where $f'(r, \theta)$ is the reconstructed image. The basis functions $R_n(r) \exp(il\theta)$ of the PHT are orthogonal over the interior of the unit circle, and each order of the PHT coefficients makes an independent contribution to the reconstruction of the image.

Figure 1 gives a example of image reconstruction using ZMs and PHT for standard gray image “Lena” (moment orders $K = 5, 10, 15, 20, 25, 30, 35, 40, 45, 50$). It can be observed that (1) the reconstructed images using PHT show more visual resemblance to the original image in the early orders and the edges of the reconstructed images are also better defined with less jaggedness; (2) the numerical stability of ZMs breaks down when the number of moments is increased to 45, and PHT gives better reconstruction results when compared to ZMs.

3. THE QUATERNION POLAR HARMONIC TRANSFORM

Let $f(r, \theta)$ be a RGB color image defined in polar coordinates, we define the right-side quaternion Polar Harmonic Transform (QPHT) of order n with repetition l as in the

works by Chen et al. [2012] and Guo et al. [2011].

$$\begin{aligned} M_{n,l}^R &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) [H_{n,l}(r, \theta)]^* r dr d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) \exp(-\mu 2\pi n r^2) \exp(-\mu l \theta) r dr d\theta, \end{aligned} \quad (12)$$

where μ is an unit pure quaternion chosen as $\mu = (\mathbf{i} + \mathbf{j} + \mathbf{k}) / \sqrt{3}$.

Since the radial Polar Harmonic basis functions are orthogonal, the color image $f(r, \theta)$ can be reconstructed approximately from a limited orders of QPHT coefficients ($n \leq n_{\max}, l \leq l_{\max}$). The more orders used, the more accurate the color image description

$$\begin{aligned} f'(r, \theta) &= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l}^R R_n(r) \exp(\mu l \theta) = \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l}^R \exp(\mu 2\pi n r^2) \exp(\mu l \theta) \\ &\approx \sum_{n=-n_{\max}}^{+n_{\max}} \sum_{l=-l_{\max}}^{+l_{\max}} M_{n,l}^R \exp(\mu 2\pi n r^2) \exp(\mu l \theta), \end{aligned} \quad (13)$$

where $f'(r, \theta)$ is the reconstructed color image. The basis functions $R_n(r) \exp(\mu l \theta)$ of the QPHT are orthogonal over the interior of the unit circle, and each order of the QPHT coefficients makes an independent contribution to the reconstruction of the color image.

It can be seen from Equation (12) that the multiplication of quaternions is not commutative, so we can also define the left-side QPHT of order n with repetition l as

$$M_{n,l}^L = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 \exp(-\mu l \theta) f(r, \theta) \exp(-\mu 2\pi n r^2) r dr d\theta. \quad (14)$$

The corresponding inverse QPHT is

$$f'(r, \theta) = \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \exp(\mu l \theta) M_{n,l}^L \exp(\mu 2\pi n r^2) \approx \sum_{n=-n_{\max}}^{+n_{\max}} \sum_{l=-l_{\max}}^{+l_{\max}} \exp(\mu l \theta) M_{n,l}^L \exp(\mu 2\pi n r^2). \quad (15)$$

In this article, the QPHT refers to right-side QPHT and denoted as “ $M_{n,l}^R$ ” for abbreviation [Li 2013].

Appendix A gives the relationship between the QPHT and the conventional PHT for a single channel of the RGB image, and Appendix B gives the relationship between the inverse QPHT and the conventional inverse PHT for a single channel of the RGB image. Appendix A and B show that the QPHT can be obtained from the PHT of each color channel.

The most direct method of testing the color image representation capability of the QPHT is via color image reconstruction. In this experiment, finite sets of QPHT coefficients of the 50 test color images are first calculated, and then, from these sets of coefficients, the color images are reconstructed according to (13). The difference between the reconstructed image and the original color image is then measured using the reconstruction error [Amayeh et al. 2011] as follows.

$$\varepsilon = \frac{\sum_x \sum_y |f'(x, y) - f(x, y)|^2}{\sum_x \sum_y |f(x, y)|^2}, \quad (16)$$

where $f(x, y)$ and $f'(x, y)$ are, respectively, the original color image and the reconstructed color image.

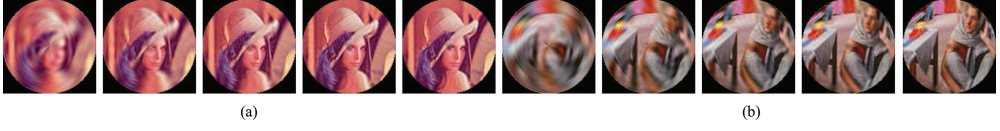


Fig. 2. The reconstructed images for color images of size 128×128 (orders $n = 10, 20, 30, 40, 50$): (a) Color image Lena, (b) Color image Barbara.

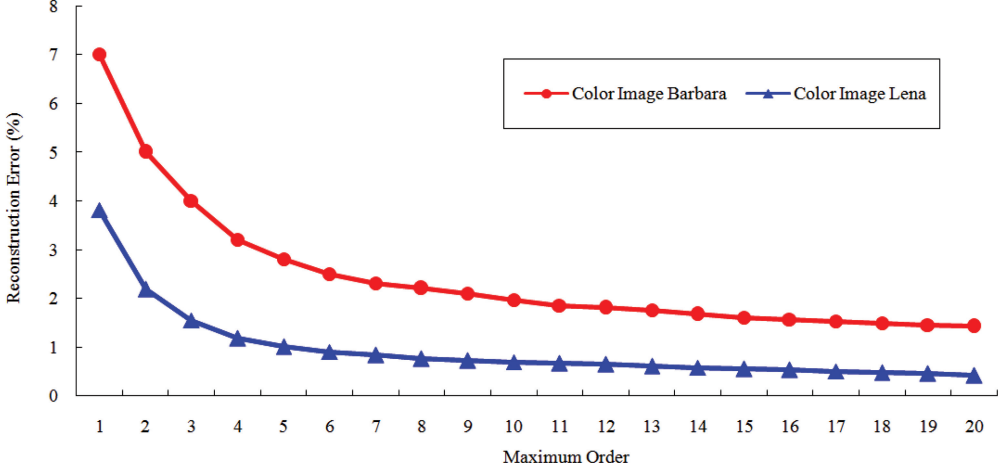


Fig. 3. The reconstruction error yielded by QPHT in the color image reconstruction experiment.

Some examples of the reconstructed color images are shown in Figure 2. As more QPHT coefficients are added to the reconstruction process, the reconstructed images get closer to the original images. As can be observed from the reconstructed images, QPHT coefficients can capture the color image information, especially the edges. In Figure 3, we show plots of the average reconstruction errors. It can be observed that the reconstructed color images using QPHT coefficients show visual resemblance to the original image in the early orders, and the QPHT is free of numerical instability issues.

4. RST INVARIANTS OF QUATERNION POLAR HARMONIC TRANSFORM

Here, we will derive and analyze the RST invariant property of QPHT [Chen et al. 2012; Guo et al. 2011; Xin et al. 2007].

4.1. Translation Invariance

Translation invariance can be achieved by putting the origin of coordinates at the color image centroid. Suk et al. [2009] have defined the common centroid (x_c, y_c) of all three color channels as follows

$$\begin{aligned}
 x_c &= (h_{1,0}(f_R) + h_{1,0}(f_G) + h_{1,0}(f_B)) / h_{0,0} \\
 y_c &= (h_{0,1}(f_R) + h_{0,1}(f_G) + h_{0,1}(f_B)) / h_{0,0} \\
 h_{0,0} &= h_{0,0}(f_R) + h_{0,0}(f_G) + h_{0,0}(f_B),
 \end{aligned} \tag{17}$$

where $h_{0,0}(f_R)$, $h_{1,0}(f_R)$, and $h_{0,1}(f_R)$ are respectively the zero-order and first-order geometric moment for R color channel, $h_{0,0}(f_G)$, $h_{1,0}(f_G)$, and $h_{0,1}(f_G)$ for G channel, and $h_{0,0}(f_B)$, $h_{1,0}(f_B)$, and $h_{0,1}(f_B)$ for B channel.

Let the origin of the coordinate system be located at (x_c, y_c) , the central QPHT, which are invariant to color image translation, can be obtained as follows:

$$\begin{aligned}\bar{M}_{n,l}^R &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(\bar{r}, \bar{\theta}) [H_{n,l}(\bar{r}, \bar{\theta})]^* \bar{r} d\bar{r} d\bar{\theta} \\ &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(\bar{r}, \bar{\theta}) \exp(-\mu 2\pi n \bar{r}^2) \exp(-\mu l \bar{\theta}) \bar{r} d\bar{r} d\bar{\theta},\end{aligned}\quad (18)$$

where $(\bar{r}, \bar{\theta})$ is the color image pixel coordinate representation in polar form by locating the origin at (x_c, y_c) .

4.2. Rotation Invariance

Let $f^r(r, \theta) = f(r, \theta + \alpha)$ denote the rotation change of a color image $f(r, \theta)$ by the angle α , then QPHT of $f(r, \theta + \alpha)$ and $f(r, \theta)$ have the following relations

$$\begin{aligned}M_{n,l}^R(f^r) &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f^r(r, \theta) R_n^*(r) \exp(-\mu l \theta) r dr d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta + \alpha) R_n^*(r) \exp(-\mu l \theta) r dr d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) R_n^*(r) \exp(-\mu l (\theta - \alpha)) r dr d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) R_n^*(r) \exp(-\mu l \theta) r dr d\theta \times \exp(\mu l \alpha) \\ &= M_{n,l}^R(f) \exp(\mu l \alpha),\end{aligned}\quad (19)$$

where $M_{n,l}^R(f^r)$ and $M_{n,l}^R(f)$ are the QPHT coefficients of $f^r(r, \theta)$ and $f(r, \theta)$, respectively.

According to Equation (19), we know that a rotation of the color image by an angle α induces a phase shift $e^{\mu l \alpha}$ of the $M_{n,l}^R(f)$. Taking the norm on both sides of Equation (19), we have

$$|M_{n,l}^R(f^r)| = |M_{n,l}^R(f) \exp(\mu l \alpha)| = |M_{n,l}^R(f)| |\exp(\mu l \alpha)| = |M_{n,l}^R(f)|. \quad (20)$$

So, the rotation invariance can be achieved by taking the norm of the color images' QPHT. In other words, the QPHT coefficients modulus $|M_{n,l}^R(f)|$ are invariant with respect to rotation transform.

4.3. Scaling Invariance

The QPHT coefficients modulus are invariant to scaling if the computation area can be made to cover the same content. In practice, this condition is met because the QPHT are defined on the unit disk. Given a color image defined on a discrete domain $f(k, l)$, where $k = 0, \dots, P-1$ and $l = 0, \dots, Q-1$, we can map the color image to a unit-disk domain of $(x_k, y_l) \in [-1, 1] \times [-1, 1]$ with

$$x_k = \frac{k - P/2}{P/2}, \quad y_l = \frac{l - Q/2}{Q/2}. \quad (21)$$

Figure 4 shows the modulus distribution of QPHT coefficients for color image Lena under various attacks. It can be seen that the QPHT coefficients modulus have good robustness against various noises, geometric transformations, and color variations. So, QPHT coefficients modulus are suitable for robust color image watermarking.

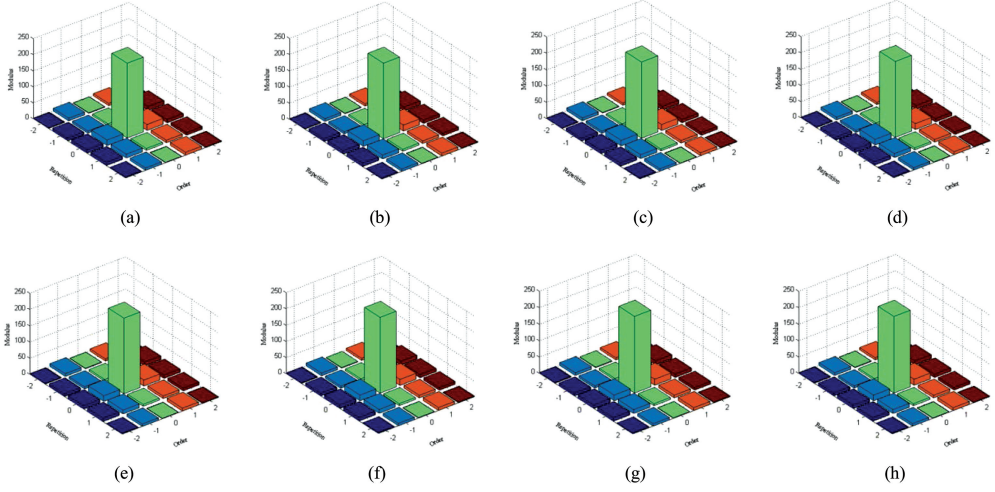


Fig. 4. The modulus distribution of QPHT coefficients for color image Lena under various attacks (moment order $K = 5$): (a) image blurring, (b) light increasing, (c) contrast lowering, (d) JPEG 50, (e) median filtering, (f) Gaussian noise, (g) rotation, (h) scaling.

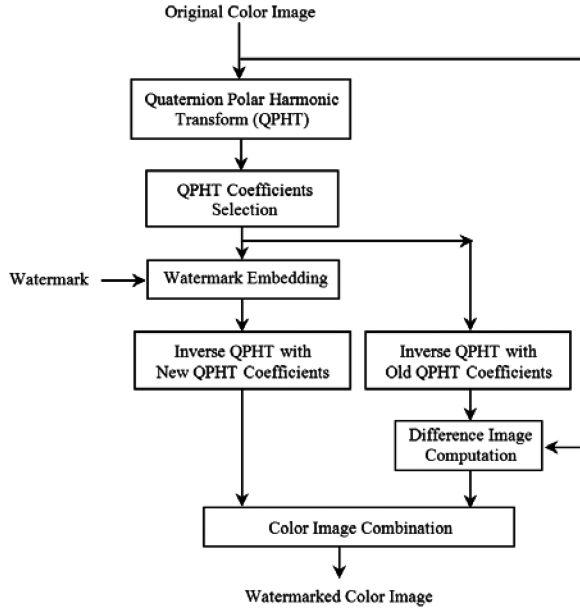


Fig. 5. Watermark embedding.

5. ROBUST COLOR IMAGE WATERMARKING APPROACH USING INVARIANT QPHT COEFFICIENTS MODULUS

In this article, we propose a robust color image watermarking approach using invariant QPHT coefficients modulus. The approach embeds the watermark information into original color image by adaptively modulating the QPHT coefficients. In the decoder, the digital watermark can be retrieved from the invariant QPHT coefficients directly. The diagram of our color image watermark embedding scheme is shown in Figure 5.

Table I. Modulus of Some QPHT Coefficients for a Color Image with Constant Color Value 60

	l = 0	l = 1	l = 2	l = 3	l = 4	l = 5	l = 6	l = 7	l = 8	l = 9	l = 10
n = 0	103.72	0.0000	0.0000	0.0000	0.2537	0.0000	0.0000	0.0000	0.1620	0.0000	0.0000
n = 1	0.2029	0.0000	0.0000	0.0000	0.2537	0.0000	0.0000	0.0000	0.1620	0.0000	0.0000
n = 2	0.2030	0.0000	0.0000	0.0000	0.2537	0.0000	0.0000	0.0000	0.1621	0.0000	0.0000
n = 3	0.2030	0.0000	0.0000	0.0000	0.2538	0.0000	0.0000	0.0000	0.1622	0.0000	0.0000
n = 4	0.2030	0.0000	0.0000	0.0000	0.2538	0.0000	0.0000	0.0000	0.1622	0.0000	0.0000
n = 5	0.2030	0.0000	0.0000	0.0000	0.2538	0.0000	0.0000	0.0000	0.1622	0.0000	0.0000
n = 6	0.2030	0.0000	0.0000	0.0000	0.2538	0.0000	0.0000	0.0000	0.1622	0.0000	0.0000
n = 7	0.2031	0.0000	0.0000	0.0000	0.2539	0.0000	0.0000	0.0000	0.1623	0.0000	0.0000
n = 8	0.2031	0.0000	0.0000	0.0000	0.2539	0.0000	0.0000	0.0000	0.1623	0.0000	0.0000
n = 9	0.2031	0.0000	0.0000	0.0000	0.2539	0.0000	0.0000	0.0000	0.1623	0.0000	0.0000
n = 10	0.2031	0.0000	0.0000	0.0000	0.2539	0.0000	0.0000	0.0000	0.1623	0.0000	0.0000

5.1. Watermark Embedding

Let $I = \{f(x, y), 0 \leq x < M, 0 \leq y < N\}$ represent a host digital image (color image), and $f(x, y)$ denotes the color value at position (x, y) .

$W = \{w(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q\}$ is a binary watermark image to be embedded within the host image. The digital watermark embedding can be summarized as follows.

Step 1: Watermark preprocessing. To ensure the security of the watermark and make the watermark be hard to be extracted, the digital watermark should be scrambled. In this work, Arnold transform is chosen as pretreatment method for watermark signal, this is because Arnold transform is intuitionistic, simple, periodic, and easy to use. Here, the binary watermark image is scrambled from W to W_1 by using the Arnold transform, where

$$W_1 = \{w_1(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q\}. \quad (22)$$

Then it is transformed into a one-dimensional sequence of ones and zeros as follows:

$$W_2 = \{w_2(k) = w_1(i, j), 0 \leq i < P, 0 \leq j < Q, k = i \times Q + j, w_2(k) \in \{0, 1\}\}.$$

Step 2: Quaternion Polar Harmonic Transform (QPHT). The QPHT is performed on the original color image I (see Section 3).

Step 3: QPHT coefficients selection. The QPHT is defined for analog images. For digital images, the QPHT can only be obtained approximately. Accordingly, the invariance of the QPHT is achieved approximately. For watermarking, only the accurate QPHT coefficients should be used [Li et al. 2012]. In order to select the accurate QPHT coefficients, we compute the QPHT coefficients of 128×128 color image with constant color value 60, and the modulus of some QPHT coefficients are listed in Table I.

From Table I, it could be concluded that the modulus of QPHT coefficients with repetition $l = 4m, m \in \mathbb{Z}$, are not zero [Xin et al. 2007]. Some of them even have significant values. This shows that the QPHT coefficients with repetitions $l = 4m, m \in \mathbb{Z}$, are not accurate, so they cannot be used to encode watermark bits. As a result, according to the reconstruction accuracy and geometric invariance of QPHT coefficients, the final moment set used for watermark embedding in the proposed scheme are as follows

$$S = \{M_{n,l}, l \neq 4m, m \in \mathbb{Z}\}.$$

Step 4: Digital watermark embedding. Once the accurate QPHT coefficients are determined, a binary watermark sequence is then embedded by modifying the modulus

of the selected QPHT coefficients. For the binary watermark sequence $W_2 = \{w_2(k) \mid 0 \leq k < P \times Q\}$, a secret key K_1 is first employed to randomly select $P \times Q$ QPHT coefficients $M(k) (0 \leq k < P \times Q)$ from the accurate QPHT coefficients set S . Then the watermark bits are embedded by modifying the modulus of the selected QPHT coefficients using the following quantization function [Li et al. 2010]:

$$|M'(k)| = \begin{cases} 2\Delta * \text{round}\left(\frac{|M(k)|}{2\Delta}\right) + \Delta/2 & \text{if } w_2(k) = 1 \\ 2\Delta * \text{round}\left(\frac{|M(k)|}{2\Delta}\right) - \Delta/2 & \text{if } w_2(k) = 0 \end{cases} \quad (0 \leq k < P \times Q), \quad (23)$$

where $|M(k)|$ is the old QPHT coefficients modulus of host image, $|M'(k)|$ is the new QPHT coefficients modulus, $\text{round}(\cdot)$ denotes round operator, and Δ is the watermark embedding strength.

Step 5: Obtaining the watermarked color image. The watermarked color image I' is composed of two parts. One part is the components contributed by those unchanged QPHT coefficients, which is

$$f_{rem}(r, \theta) = f(r, \theta) - f_M(r, \theta), \quad (24)$$

where the second term is the components contributed by the selected QPHT coefficients before they are changed:

$$f_M(\cdot) = \sum_{i=0}^{P \times Q - 1} M_{n_i, l_i} H_{n_i, l_i}(\cdot) + M_{n_i, -l_i} H_{n_i, -l_i}(\cdot). \quad (25)$$

The other part is the components contributed by those modified QPHT coefficients:

$$f_{M'}(\cdot) = \sum_{i=0}^{P \times Q - 1} M'_{n_i, l_i} H_{n_i, l_i}(\cdot) + M'_{n_i, -l_i} H_{n_i, -l_i}(\cdot). \quad (26)$$

Therefore we obtain the watermarked color image I' by combining the two parts

$$f'(r, \theta) = f_{rem}(r, \theta) + f_{M'}(r, \theta). \quad (27)$$

5.2. Watermark Extraction

The watermark extraction procedure in the proposed scheme neither needs the original image nor any other side information. Let I^* denote the watermarked color image, the main steps of watermark extraction can be described as follows.

Step 1: Quaternion Polar Harmonic Transform. The QPHT is performed on the watermarked color image I^* (see Section 3).

Step 2: Accurate QPHT coefficients selection. With the same key K_1 as in the process of watermark insertion, we locate the QPHT coefficients $M^*(k) (0 \leq k < P \times Q)$ that carry the watermark information.

Step 3: Digital watermark extraction. The one-dimensional binary sequence is extracted from the modulus of the selected QPHT coefficients

$$w^*(k) = \begin{cases} 1 & \text{if } |M^*(k)| - 2\Delta * \text{round}\left(\frac{|M^*(k)|}{2\Delta}\right) > 0 \\ 0 & \text{if } |M^*(k)| - 2\Delta * \text{round}\left(\frac{|M^*(k)|}{2\Delta}\right) \leq 0 \end{cases} \quad (0 \leq k < P \times Q), \quad (28)$$

where $W^* = \{w^*(k), 0 \leq k < P \times Q\}$ is the extracted one-dimensional binary sequence.

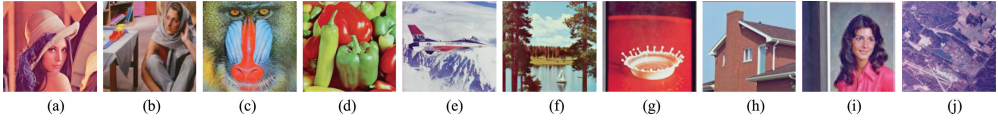


Fig. 6. The host color images used in experiments: (a) Lena, (b) Barbara, (c) Mandrill, (d) Peppers, (e) Airplane, (f) Sailboat and lake, (g) Splash, (h) House, (i) Girl, (j) San Diego.

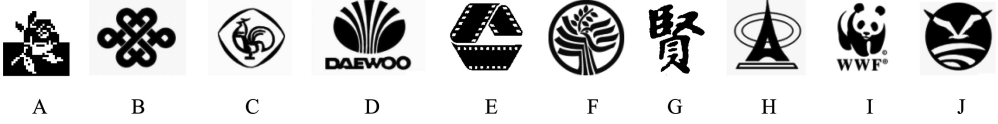


Fig. 7. Digital watermarks A through J used in experiments.

Step 4: Obtaining the binary watermark image. The one-dimensional binary sequences W^* is rearranged to form the binary image, and the binary watermark image

$$\hat{W} = \{\hat{w}(i, j), 0 \leq i < P, 0 \leq j < Q\} \quad (29)$$

can be obtained by descrambling.

6. SIMULATION RESULTS

We tested the proposed watermarking scheme on ten popular test images. Ten test images from USC-SIPI image database, each of dimensions $256 \times 256 \times 24\text{bit}$, are shown in Figure 6, referred to as “Lena,” “Barbara,” “Mandrill,” “Peppers,” “Airplane,” “Sailboat and lake,” “Splash,” “House,” “Girl,” and “San Diego,” respectively, in the sequel. And ten test watermarks, each of dimensions 16×16 , are shown in Figure 7, hereinafter referred to as watermarks A, B, C, D, E, F, G, H, I, and J, respectively. The embedding strength $\Delta = 0.3$. Also, the experimental results are compared with schemes in [Chou et al. 2010; Niu et al. 2011; Tsui et al. 2008]. All the experiments are carried out on a Pentium 4 PC with 2 GHz CPU and 1 GB memory, within MATLAB 7.0 in Windows XP.

- (1) Chou et al.’s scheme is a color image watermarking scheme that hides watermark signals in most distortion-tolerable signals within three color channels of the host image [Chou et al. 2010]. To reinforce the robustness, the watermark signals are repeated and permuted before embedding, and restored by the majority-vote decision making process in watermark extraction.
- (2) Niu et al.’s algorithm is a color image watermarking algorithm based on the *support vector regression* (SVR) and *nonsubsampling contourlet transform* (NSCT) [Niu et al. 2011]. The digital watermark is embedded into the green component of color images by modifying the selected NSCT coefficient blocks. In watermark detection, according to the high correlation among different channels of the color image, the watermark be recovered by using SVR technique.
- (3) Tsui et al.’s approach use a color image watermarking approach that encodes the components of color images and watermarks are embedded as vectors in the frequency domain of the channels by using the quaternion Fourier transform [Tsui et al. 2008]. Robustness is achieved by embedding a watermark in the coefficient with positive frequency, which spreads it to all color components in the spatial domain and invisibility is satisfied by modifying the coefficient with negative frequency, such that the combined effects of the two are insensitive to human eyes.

In this work, the watermarking performance is evaluated from two aspects, namely invisibility and robustness. Here, we use *peak signal-to-noise ratio* (PSNR) to measure the watermark invisibility, and watermark robustness is measured as the *bit error rate* (BER) of extracted watermark.

Each of the ten test watermarks was embedded in the ten test images using the proposed watermarking method. A total of $10 \times 10 = 100$ watermarked color images were generated and for each watermarked color image, the digital watermark was extracted under various attacks. Table II gives the average values of the PSNRs obtained after embedding a particular watermark in the ten test images, as well as the average values of the BER obtained under various attacks.

6.1. Watermark Invisibility

Figure 8 is the result of applying our algorithm for data embedding in color images. Figure 8(a) show the watermarked image Lena, Barbara, Mandrill, and Peppers using our algorithm. Figure 8(b) are the absolute difference between origin color image and watermarked image for our algorithm respectively, multiplied by 50 for better display. From Figure 8, we can see that the watermarked color images have good subjective quality. Table III gives the PSNR of different watermarking schemes. From Table III, we can know that the proposed scheme is better than schemes by Chou et al. [2010], Niu et al. [2011], and Tsui et al. [2008] in terms of the transparency.

6.2. Robustness to Various Attacks

Some simulation results, which are obtained by the proposed watermarking scheme, for common image processing operations and geometric distortions are shown in Figure 9. Table IV and Table V show the results of comparison with schemes by Chou et al. [2010], Niu et al. [2011], and Tsui et al. [2008].

It is clear that the proposed scheme outperforms the schemes by Chou et al. [2010], Niu et al. [2011], and Tsui et al. [2008] under most attacks in terms of BER.

From here measurements, several observations can be made.

- The proposed color image watermarking method. Our watermarking method has good imperceptibility, for the kernel computation of QPHT has no numerical stability issue. This method provides satisfactory results for most attacks, especially for color attacks and geometric distortions. But it is necessary to point out that, for the proposed watermarking method, the resistance against local image cropping is not ideal (as shown in Figure 10), and this is because QPHT is global statistical based image feature extraction.
- The perceptually tuned color image watermarking [Chou et al. 2010]. The scheme has better imperceptibility than Tsui et al.'s approach [Tsui et al. 2008], because it is perceptually optimized by embedding high-strength watermark signals in wavelet coefficients of high perceptual redundancy. The strength of the embedded watermark signal is determined by the amount of perceptual redundancy inherent in the corresponding wavelet coefficient. However, it is fragile to some color attacks and geometric distortions, including rotation, translation etc. The underlying reason is that it ignores the color channel correlation and the watermark desynchronization.
- The NSCT domain color image watermarking algorithm [Niu et al. 2011]. This algorithm provides good watermark imperceptibility, and satisfactory robustness to most attacks. But, it is not robust to some color attacks, such as image blurring, light increasing, and contrast lowering etc. The underlying reason is that it only embeds the digital watermark into one image color channel, and ignores the correlation between different color channels. Another disadvantage of it is that this algorithm

Table II. The Average Performance of Different Watermarking Method

Watermark	Methods	Average PSNR (No Attack)	Average BER							
			Median Filtering (3 × 3)	Average Filtering (3 × 3)	Salt and Peppers Noise (0.01)	Gaussian Noise (0.01)	JPEG 50	Rotation 15°	Scaling 1.5	Translation (H 2,V 15)
A	Proposed Scheme	48.23dB	0.0083	0.0202	0.0045	0.0102	0.0112	0.0016	0.0112	0.0006
	Chou et al. 2010	40.91dB	0.0502	0.0468	0.0713	0.1986	0.0756	N/A	0.0336	N/A
	Niu et al. 2011	41.54dB	0.0270	0.0311	0.0225	0.0253	0.0332	0.0358	0.0275	0.0351
	Tsui et al. 2008	34.63dB	0.0122	0.0180	0.0096	0.0091	0.0106	N/A	0.0413	N/A
B	Proposed Scheme	48.52dB	0.0163	0.0198	0.0032	0.0112	0.0107	0.0003	0.0074	0.0005
	Chou et al. 2010	41.03dB	0.0520	0.0458	0.0718	0.2023	0.0710	N/A	0.0358	N/A
	Niu et al. 2011	41.63dB	0.0258	0.0298	0.0231	0.02598	0.0340	0.0363	0.0277	0.0318
	Tsui et al. 2008	34.51dB	0.0201	0.0131	0.0103	0.0099	0.0124	N/A	0.0416	N/A
C	Proposed Scheme	47.81dB	0.0126	0.0217	0.0056	0.0064	0.0124	0.0016	0.0098	0.0012
	Chou et al. 2010	40.93dB	0.0523	0.0457	0.0752	0.2165	0.0747	N/A	0.0371	N/A
	Niu et al. 2011	41.80dB	0.0261	0.0311	0.0236	0.0269	0.0331	0.0370	0.0288	0.0318
	Tsui et al. 2008	34.65dB	0.0158	0.0177	0.0124	0.0113	0.0201	N/A	0.0414	N/A
D	Proposed Scheme	48.37dB	0.0134	0.0223	0.0027	0.0102	0.0064	0.0006	0.0042	0.0021
	Chou et al. 2010	41.21dB	0.0511	0.0480	0.0747	0.2168	0.0748	N/A	0.0368	N/A
	Niu et al. 2011	41.63dB	0.0257	0.0314	0.0230	0.0268	0.0325	0.0375	0.0288	0.0315
	Tsui et al. 2008	34.67dB	0.0132	0.0190	0.0135	0.0105	0.0173	N/A	0.0403	N/A
E	Proposed Scheme	48.15dB	0.0156	0.0173	0.0095	0.0056	0.0108	0.0017	0.0156	0.0018
	Chou et al. 2010	40.92dB	0.0531	0.0468	0.0751	0.2128	0.0770	N/A	0.0371	N/A
	Niu et al. 2011	41.80dB	0.0261	0.0324	0.0227	0.0280	0.0332	0.0368	0.0275	0.0317
	Tsui et al. 2008	34.62dB	0.0162	0.0194	0.0114	0.0117	0.0156	N/A	0.0415	N/A

(Continued)

Table II. Continued

Watermark	Methods	Average PSNR (No Attack)	Average BER							
			Median Filtering (3 × 3)	Average Filtering (3 × 3)	Salt and Peppers Noise (0.01)	Gaussian Noise (0.01)	JPEG 50	Rotation 15°	Scaling 1.5	Translation (H 2,V 15)
F	Proposed Scheme	48.18dB	0.0157	0.0205	0.0065	0.0073	0.0096	0.0008	0.0072	0.0010
	Chou et al. 2010	40.64dB	0.0521	0.0473	0.0762	0.2154	0.0771	N/A	0.0358	N/A
	Niu et al. 2011	41.85dB	0.0270	0.0313	0.0240	0.0268	0.0326	0.0369	0.0280	0.0317
	Tsui et al. 2008	34.47dB	0.0156	0.0188	0.0116	0.0114	0.0113	N/A	0.0431	N/A
G	Proposed Scheme	47.90dB	0.0078	0.0198	0.0033	0.0165	0.0115	0.0011	0.0074	0.0008
	Chou et al. 2010	41.13dB	0.0525	0.0448	0.0737	0.2041	0.0747	N/A	0.0335	N/A
	Niu et al. 2011	41.79dB	0.0253	0.0312	0.0240	0.0268	0.0342	0.0358	0.0290	0.0316
	Tsui et al. 2008	34.54dB	0.0124	0.0175	0.0124	0.0133	0.0163	N/A	0.0423	N/A
H	Proposed Scheme	47.73dB	0.0143	0.0192	0.0050	0.0124	0.0183	0.0028	0.0155	0.0006
	Chou et al. 2010	40.92dB	0.0534	0.0469	0.0726	0.2026	0.0753	N/A	0.0368	N/A
	Niu et al. 2011	41.63dB	0.0252	0.0312	0.0240	0.0258	0.0342	0.0380	0.0285	0.0335
	Tsui et al. 2008	34.62dB	0.0152	0.0180	0.0116	0.0172	0.0125	N/A	0.0407	N/A
I	Proposed Scheme	48.59dB	0.0135	0.0168	0.0021	0.0101	0.0046	0.0015	0.0143	0.0004
	Chou et al. 2010	41.25dB	0.0505	0.0457	0.0737	0.2035	0.0770	N/A	0.0371	N/A
	Niu et al. 2011	41.85dB	0.0270	0.0312	0.0240	0.0267	0.0327	0.0372	0.0288	0.0331
	Tsui et al. 2008	34.67dB	0.0153	0.0180	0.0145	0.0135	0.0169	N/A	0.0405	N/A
J	Proposed Scheme	48.35dB	0.0094	0.0184	0.0048	0.0106	0.0039	0.0004	0.0086	0.0011
	Chou et al. 2010	41.17dB	0.0506	0.0468	0.0746	0.2104	0.0780	N/A	0.0359	N/A
	Niu et al. 2011	41.86dB	0.0258	0.0310	0.0227	0.0271	0.0325	0.0373	0.0278	0.0326
	Tsui et al. 2008	34.57dB	0.0184	0.0192	0.0120	0.0115	0.0172	N/A	0.0409	N/A



Fig. 8. The watermark embedding examples by using our algorithm: (a) the watermarked color image, (b) the absolute difference between original image and watermarked image.

Table III. The PSNR for Various Watermarking Methods

Image	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008
Lena	48.49	41.32	41.86	34.60
Barbara	48.21	41.18	41.96	34.67
Mandrill	47.92	40.69	42.68	33.91
Peppers	48.36	40.90	41.88	34.87



Fig. 9. The attacked watermarked image and the extracted watermark: (a) median filtering(3*3), BER = 0.0078, (b) Gaussian noise(0.01), BER = 0.0117, (c) JPEG 30, BER = 0.0156, (d) Image blurring(4,20), BER = 0.0078, (e) light increasing, BER = 0.0195, (f) Mosaic(2), BER = 0, (g) rotation (15°), BER = 0, (h) scaling (0.5), BER = 0.0117.

has high computational complexity for SVR training, NSCT transform and image normalization.

—The multidimensional Fourier transform based color image watermarking approach [Tsui et al. 2008]. This color image watermarking approach has low watermark imperceptibility, because it does not modify the symmetric multidimensional Fourier transform coefficients. In addition, this approach is fragile to some geometric distortions, including rotation, translation etc. The underlying reason is that it also ignores the watermark desynchronization.

In sum, the experimental results demonstrate the proposed color image watermarking is robust to most common image processing operations (including color attacks) and geometric distortions, and has good imperceptibility. This is because (1) QPHT has excellent color image representation capability, and is free of numerical instability issues. As more moments are added to the reconstruction process, the QPHT reconstructed images get closer to the original images. The QPHT can capture effectively

Table IV. The Watermark Detection Results for Common Image Processing Operations (BER & PSNR)

Attacks	Lena				Barbara				Mandrill			
	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008
No attack	0	0.0078	0.0117	0	0	0.0078	0.0156	0.0039	0	0.0117	0	0.0039
	48.49dB	41.24dB	41.63dB	34.47dB	48.21dB	41.17dB	41.90dB	34.26dB	47.92dB	40.87dB	42.80dB	33.78dB
	0.0078	0.0430	0.0234	0.0469	0.0156	0.0625	0.0234	0.0469	0.0117	0.0626	0.0273	0.0547
	44.65dB	36.98dB	38.14dB	31.62dB	44.52dB	37.67dB	38.63dB	31.24dB	43.44dB	37.16dB	39.31dB	30.56dB
Median Filtering (3×3)	0	0.0352	0.0234	0.0078	0.0078	0.0352	0.0195	0.0625	0.0117	0.0430	0.0117	0.0586
	42.36dB	35.81dB	36.68dB	29.23dB	43.36dB	36.25dB	37.90dB	29.95dB	42.02dB	35.82dB	37.63dB	28.43dB
	0.0195	0.0390	0.0273	0.0508	0.0156	0.0625	0.0234	0.0195	0.0195	0.0781	0.0273	0.0195
	43.64dB	35.93dB	37.34dB	30.43dB	43.68dB	36.41dB	36.97dB	28.91dB	42.94dB	36.54dB	37.80dB	28.97dB
Gaussian Noise (0.01)	0.0117	0.0781	0.0195	0.0469	0.0078	0.0742	0.0195	0.0469	0.0078	0.0781	0.0195	0.0508
	40.45dB	32.17dB	32.90dB	28.67dB	39.72dB	31.68dB	32.71dB	28.76dB	38.76dB	32.55dB	34.78dB	28.24dB
	0	0.0664	0.0195	0.0469	0.0039	0.0625	0.0156	0.0703	0.0039	0.0703	0.0195	0.0781
	39.54dB	32.80dB	32.32dB	27.41dB	38.40dB	31.50dB	31.73dB	27.50dB	38.24dB	30.62dB	32.65dB	26.91dB
Random Noise (10)	0	0.0664	0.0234	0.0703	0.0039	0.0781	0.0156	0.0469	0.0039	0.0781	0.0039	0.0508
	46.50dB	38.90dB	39.73dB	32.62dB	46.35dB	39.52dB	40.11dB	33.41dB	45.61dB	39.56dB	42.03dB	32.23dB
	0	0.0273	0.0234	0.0313	0	0.0234	0.0195	0.0352	0	0.0352	0.0195	0.0352
	46.43B	39.14dB	39.81dB	33.83dB	46.56dB	39.23dB	39.50dB	33.74dB	45.72dB	38.90dB	41.32dB	31.91dB
JPEG	0.0039	0.0469	0.0273	0.0430	0.0039	0.0547	0.0234	0.0430	0.0078	0.0547	0.0195	0.0430
	45.13dB	38.33dB	38.40dB	32.87dB	45.03dB	37.89dB	38.50dB	32.36dB	44.72dB	37.29dB	39.77dB	30.86dB
	0.0039	0.0547	0.0313	0.0508	0.0078	0.0625	0.0273	0.0547	0.0117	0.0586	0.0234	0.0547
	44.52dB	37.16dB	37.62dB	31.40dB	44.15dB	36.22dB	37.91dB	31.33dB	43.46dB	36.41dB	38.50dB	29.68dB
40	0.0078	0.0625	0.0352	0.0703	0.0117	0.0703	0.0273	0.0625	0.0156	0.0664	0.0352	0.0664
	43.25dB	35.90dB	36.76dB	30.47dB	43.24dB	34.59dB	36.78dB	30.50dB	42.25dB	35.32dB	36.93dB	28.41dB
	0.0156	0.0703	0.0390	0.0742	0.0273	0.0781	0.0313	0.0742	0.0351	0.0742	0.0390	0.0703
	42.16dB	33.50dB	35.34dB	29.42dB	41.80dB	33.36dB	34.14dB	29.32dB	40.61dB	33.78dB	34.92dB	27.35dB

(Continued)

Table IV. Continued

	Lena				Barbara				Mandrill			
	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008
Attacks	Image Blurring	0.0078	0.0195	0.0352	0.0234	0.0078	0.0352	0.0273	0.0117	0.0469	0.0117	0.0430
		42.12dB	34.47dB	36.33dB	28.25dB	42.54dB	34.41dB	28.92dB	41.96dB	34.35dB	36.65dB	28.90dB
		0.0195	0.0508	0.0234	0.0234	0.0156	0.0430	0.0195	0.0117	0.0469	0.0117	0.0273
Light Increasing	Light Increasing	43.51dB	36.17dB	37.40dB	28.63dB	43.67dB	35.77dB	28.38dB	42.28dB	35.80dB	36.24dB	28.35dB
		0.0195	0.0625	0.0273	0.0078	0.0117	0.0664	0.0195	0.0156	0.0703	0.0195	0.0586
		42.90dB	35.57dB	36.80dB	28.94dB	43.56dB	35.56dB	27.91dB	42.64dB	35.30dB	36.43dB	28.23dB
Contrast Increasing	Contrast Increasing	0.0156	0.0508	0.0195	0.0195	0.0117	0.0664	0.0156	0.0156	0.0625	0.0313	0.0234
		43.10dB	36.77dB	37.83dB	28.54dB	44.45dB	36.50dB	28.37dB	42.84dB	35.88dB	37.41dB	28.69B
		0.0195	0.0664	0.0195	0.0195	0.0078	0.0742	0.0117	0.0117	0.0742	0.0313	0.0234
Contrast Lowering	Contrast Lowering	43.55dB	36.58dB	36.90dB	28.89dB	44.61dB	36.52dB	28.23dB	44.01dB	36.30dB	37.83dB	28.38dB
		0	0.0781	0.0273	0.0508	0.0078	0.0781	0.0742	0.0156	0.0781	0.0195	0.0781
		41.30dB	32.61dB	34.69dB	26.98dB	42.34dB	33.31dB	27.78dB	40.67dB	33.26dB	35.79dB	27.40dB
Median Filtering (3 × 3) + JPEG 70	Median Filtering (3 × 3) + JPEG 70	0.0078	0.0508	0.0313	0.0508	0.0156	0.0664	0.0508	0.0117	0.0703	0.0234	0.0586
		41.24dB	33.60dB	34.82dB	26.76dB	41.85dB	33.52dB	27.32dB	39.83dB	32.90dB	35.44dB	27.35dB
		0.0078	0.0703	0.0390	0.0508	0.0078	0.0664	0.0703	0.0117	0.0742	0.0273	0.0781
Salt and Peppers Noise (0.01) + Gaussian Filtering (3 × 3) + JPEG 70	Salt and Peppers Noise (0.01) + Gaussian Filtering (3 × 3) + JPEG 70	41.12dB	33.11dB	34.01dB	26.68dB	41.54dB	33.72dB	27.12dB	40.45dB	33.21dB	35.69dB	27.30dB

Table V. The Watermark Detection Results for Geometric Distortions (BER)

Attacks	Lena				Barbara				Mandrill			
	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008	Proposed Scheme	Chou et al. 2010	Niu et al. 2011	Tsui et al. 2008
Rotation	5°	0	N/A	0.0273	N/A	0	0.0234	N/A	0	N/A	0.0156	N/A
	15°	0	N/A	0.0352	N/A	0	0.0273	N/A	0	N/A	0.0195	N/A
	45°	0	N/A	0.0313	N/A	0	0.0195	N/A	0	N/A	0.0117	N/A
	70°	0	N/A	0.0273	N/A	0	0.0547	N/A	0	N/A	0.0117	N/A
	90°	0	N/A	0.0234	N/A	0	0.0273	N/A	0	N/A	0.0195	N/A
Scaling	0.5	0.0117	0.0352	0.0273	0.0430	0.0156	0.0313	0.0390	0.0117	0.0547	0.0117	0.0390
	0.9	0	0.0234	0.0273	0.0313	0.0078	0.0352	0.0313	0.0078	0.0352	0.0078	0.0313
	1.2	0	0.0195	0.0313	0.0313	0.0273	0.0195	0.0352	0.0273	0.0273	0.0156	0.0313
	1.5	0	0.0352	0.0273	0.0390	0.0078	0.0195	0.0390	0.0039	0.0273	0.0117	0.0352
	2	0	0.0078	0.0234	0.0430	0	0.0313	0.0430	0	0.0313	0.0117	0.0430
Translation	(H 2,V 15)	0	N/A	0.0273	N/A	0	N/A	N/A	0	N/A	0.0156	N/A
	(H 20,V 20)	0	N/A	0.0234	N/A	0	N/A	N/A	0	N/A	0.0156	N/A
	(H 15,V 2)	0	N/A	0.0313	N/A	0	N/A	N/A	0	N/A	0.0195	N/A
	(H 50,V 0)	0	N/A	0.0234	N/A	0	N/A	N/A	0	N/A	0.0156	N/A
	(H 0,V 50)	0	N/A	0.0234	N/A	0	N/A	N/A	0	N/A	0.0156	N/A
Scaling 2.0 + Translation (H 0,V 15)	0.0078	N/A	0.0313	N/A	0.0039	N/A	0.0352	N/A	0.0112	N/A	0.0273	N/A
Rotation 45° + Translation (H 20,V 20)	0.0039	N/A	0.0469	N/A	0.0078	N/A	0.0547	N/A	0.0078	N/A	0.0352	N/A
Scaling 0.9 + Translation (H 5,V 15)	0.0039	N/A	0.0352	N/A	0.0112	N/A	0.0508	N/A	0.0156	N/A	0.0469	N/A
Scaling 1.2 + Translation (H 5,V 15)	0.0078	N/A	0.0508	N/A	0.0156	N/A	0.0547	N/A	0.0247	N/A	0.0547	N/A
Translation (H 5,V 15) + Salt and Peppers Noise (0.01)	0.0039	N/A	0.0547	N/A	0.0078	N/A	0.0547	N/A	0.0112	N/A	0.0469	N/A
Translation (H 20,V 20) + Gaussian Filtering (3 × 3)	0.0078	N/A	0.0390	N/A	0.0112	N/A	0.0430	N/A	0.0078	N/A	0.0507	N/A

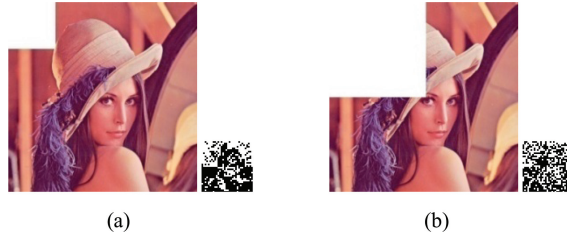


Fig. 10. The attacked (local image cropping) watermarked image and the extracted watermark: (a) cropping off (x25%, y25%), BER = 0.1172, (b) cropping off (x50%, y50%), BER = 0.2031.

the color image information, especially the edges. (2) QPHT coefficients modulus are robust to various image geometric transformations under noisy conditions. (3) We build the QPHT based color image watermarking, in which the color image is treated as a vector field and processed directly (without losing color information).

7. CONCLUSION

Geometric distortions that cause displacement between embedding and detection are usually difficult for watermark to survive. It is a challenging work to design a robust image watermarking scheme against geometric distortions, especially for color host images. In this article, we propose a robust color image watermarking scheme using QPHT. The novelty of the proposed algorithm includes: (1) QPHT is introduced, and the invariant property of QPHT is analyzed and discussed, (2) the accurate QPHT coefficients selection strategy is discussed, and (3) A new QPHT-based robust color image watermarking approach is proposed. The extensive experimental works have shown that the proposed color image watermarking strategy has conquered those challenging geometric distortions, such as rotation, translation, scaling, etc. Also, the digital watermark can resist some common image processing operations, especially including color attacks.

The deficiency of the proposed image watermarking is that it cannot effectively deal with local image cropping, this is because QPHT are global statistical based image feature extraction. Future work will focus on eliminating these drawbacks.

APPENDIX A

Let f_R , f_G , and f_B represent the classical red, green, and blue components of a color image, respectively, let $M_{n,l}(f_R)$, $M_{n,l}(f_B)$, and $M_{n,l}(f_G)$ be, respectively, the conventional PHT for three color components, and let $real(p)$ represent the real part of complex number p and $imag(p)$ be the imaginary part.

Substituting Equation (4) into Equation (12) and using Equation (2), we can obtain

$$\begin{aligned}
 M_{n,l}^R &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) [H_{n,l}(r, \theta)]^* r dr d\theta \\
 &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) \exp(-\mu 2\pi n r^2) \exp(-\mu l \theta) r dr d\theta \\
 &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) \exp(-\mu(2\pi n r^2 + l \theta)) r dr d\theta \\
 &= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 [f_R(r, \theta)i + f_G(r, \theta)j + f_B(r, \theta)k] \exp(-\mu(2\pi n r^2 + l \theta)) r dr d\theta
 \end{aligned}$$

$$\begin{aligned}
&= i \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_R(r, \theta) \exp(-\mu(2\pi nr^2 + l\theta)) r dr d\theta \\
&\quad + j \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_G(r, \theta) \exp(-\mu(2\pi nr^2 + l\theta)) r dr d\theta \\
&\quad + k \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_B(r, \theta) \exp(-\mu(2\pi nr^2 + l\theta)) r dr d\theta \\
&= i \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_R(r, \theta) (\cos(2\pi nr^2 + l\theta) - \mu \sin(2\pi nr^2 + l\theta)) r dr d\theta \\
&\quad + j \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_G(r, \theta) (\cos(2\pi nr^2 + l\theta) - \mu \sin(2\pi nr^2 + l\theta)) r dr d\theta \\
&\quad + k \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_B(r, \theta) (\cos(2\pi nr^2 + l\theta) - \mu \sin(2\pi nr^2 + l\theta)) r dr d\theta \\
&= i \left[\frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_R(r, \theta) \cos(2\pi nr^2 + l\theta) r dr d\theta \right. \\
&\quad \left. - \mu \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_R(r, \theta) \sin(2\pi nr^2 + l\theta) r dr d\theta \right] \\
&\quad + j \left[\frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_G(r, \theta) \cos(2\pi nr^2 + l\theta) r dr d\theta \right. \\
&\quad \left. - \mu \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_G(r, \theta) \sin(2\pi nr^2 + l\theta) r dr d\theta \right] \\
&\quad + k \left[\frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_B(r, \theta) \cos(2\pi nr^2 + l\theta) r dr d\theta \right. \\
&\quad \left. - \mu \frac{1}{\pi} \int_0^{2\pi} \int_0^1 f_B(r, \theta) \sin(2\pi nr^2 + l\theta) r dr d\theta \right] \\
&= i \left[\text{real}(M_{n,l}(f_R)) + \frac{i+j+k}{\sqrt{3}} \text{imag}(M_{n,l}(f_R)) \right] \\
&\quad + j \left[\text{real}(M_{n,l}(f_G)) + \frac{i+j+k}{\sqrt{3}} \text{imag}(M_{n,l}(f_G)) \right] \\
&\quad + k \left[\text{real}(M_{n,l}(f_B)) + \frac{i+j+k}{\sqrt{3}} \text{imag}(M_{n,l}(f_B)) \right] \\
&= A_{n,l}^R + iB_{n,l}^R + jC_{n,l}^R + kD_{n,l}^R,
\end{aligned}$$

where

$$A_{n,l}^R = -\frac{1}{\sqrt{3}} [\text{imag}(M_{n,l}(f_R)) + \text{imag}(M_{n,l}(f_G)) + \text{imag}(M_{n,l}(f_B))]$$

$$\begin{aligned}
B_{n,l}^R &= \text{real}(M_{n,l}(f_R)) + \frac{1}{\sqrt{3}}[\text{imag}(M_{n,l}(f_G)) - \text{imag}(M_{n,l}(f_B))] \\
C_{n,l}^R &= \text{real}(M_{n,l}(f_G)) + \frac{1}{\sqrt{3}}[\text{imag}(M_{n,l}(f_B)) - \text{imag}(M_{n,l}(f_R))] \\
D_{n,l}^R &= \text{real}(M_{n,l}(f_B)) + \frac{1}{\sqrt{3}}[\text{imag}(M_{n,l}(f_R)) - \text{imag}(M_{n,l}(f_G))].
\end{aligned}$$

APPENDIX B

According to the quaternion Fourier transform and inverse quaternion Fourier transform, the inverse QPHT can be described as follows

$$\begin{aligned}
f(r, \theta) &= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l}^R R_n(r) \exp(\mu l \theta) \\
&= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l}^R \exp(\mu 2\pi n r^2) \exp(\mu l \theta) \\
&= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} M_{n,l}^R \exp(\mu(2\pi n r^2 + l \theta)) \\
&= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} (A_{n,l}^R + i B_{n,l}^R + j C_{n,l}^R + k D_{n,l}^R) \exp(\mu(2\pi n r^2 + l \theta)) \\
&= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} A_{n,l}^R \exp(\mu(2\pi n r^2 + l \theta)) + i \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} B_{n,l}^R \exp(\mu(2\pi n r^2 + l \theta)) \\
&\quad + j \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} C_{n,l}^R \exp(\mu(2\pi n r^2 + l \theta)) + k \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} D_{n,l}^R \exp(\mu(2\pi n r^2 + l \theta)) \\
&= \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} A_{n,l}^R \exp\left(\frac{i+j+k}{\sqrt{3}}(2\pi n r^2 + l \theta)\right) \\
&\quad + i \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} B_{n,l}^R \exp\left(\frac{i+j+k}{\sqrt{3}}(2\pi n r^2 + l \theta)\right) \\
&\quad + j \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} C_{n,l}^R \exp\left(\frac{i+j+k}{\sqrt{3}}(2\pi n r^2 + l \theta)\right) \\
&\quad + k \sum_{n=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} D_{n,l}^R \exp\left(\frac{i+j+k}{\sqrt{3}}(2\pi n r^2 + l \theta)\right) \\
&= f_A'(r, \theta) + i f_B'(r, \theta) + j f_C'(r, \theta) + k f_D'(r, \theta),
\end{aligned}$$

where

$$\begin{aligned}
f_A'(r, \theta) &= \text{real}(A_{n,l}'^R) - \frac{1}{\sqrt{3}}[\text{imag}(B_{n,l}'^R) + \text{imag}(C_{n,l}'^R) + \text{imag}(D_{n,l}'^R)] \\
f_B'(r, \theta) &= \text{real}(B_{n,l}'^R) + \frac{1}{\sqrt{3}}[\text{imag}(A_{n,l}'^R) + \text{imag}(C_{n,l}'^R) - \text{imag}(D_{n,l}'^R)]
\end{aligned}$$

$$f'_C(r, \theta) = \text{real}(C'_{n,l}{}^R) + \frac{1}{\sqrt{3}}[\text{imag}(A'_{n,l}{}^R) - \text{imag}(B'_{n,l}{}^R) + \text{imag}(D'_{n,l}{}^R)]$$

$$f'_D(r, \theta) = \text{real}(D'_{n,l}{}^R) + \frac{1}{\sqrt{3}}[\text{imag}(A'_{n,l}{}^R) + \text{imag}(B'_{n,l}{}^R) - \text{imag}(C'_{n,l}{}^R)].$$

Here, $f'_A(r, \theta)$ is a matrix close to 0, $f'_B(r, \theta)$, $f'_C(r, \theta)$, and $f'_D(r, \theta)$ represent the red, green, and blue components of the reconstructed color image respectively, and $A'_{n,l}{}^R$, $B'_{n,l}{}^R$, $C'_{n,l}{}^R$, and $D'_{n,l}{}^R$ represent the reconstruction matrix of $A_{n,l}^R$, $B_{n,l}^R$, $C_{n,l}^R$, and $D_{n,l}^R$.

$$A'_{n,l}{}^R = \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} A_{n,l}^R R_n(r) \exp(il\theta)$$

$$B'_{n,l}{}^R = \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} B_{n,l}^R R_n(r) \exp(il\theta)$$

$$C'_{n,l}{}^R = \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} C_{n,l}^R R_n(r) \exp(il\theta)$$

$$D'_{n,l}{}^R = \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} D_{n,l}^R R_n(r) \exp(il\theta).$$

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